MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG3DSESTA200- Statistical Techniques for Data Science and Machine Learning (Data Analytics Specialization)

Duration: 1.5 Hours Maximum: 50 Marks

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions

Answer any seven questions. Each question carries 2 marks.

1. Define sampling distribution. [K][1]
2. Explain the two types of errors in testing a statistical hypothesis. [U][1]
3. Write any two uses of the chi-square test. [K][2]
4. What are the assumptions of one-way ANOVA? [U][2]
5. Write any two applications of Machine learning [K][3]
6. State the general form of a simple linear regression model. [K][3]
7. Define logistic regression. [K][3]
8. What is categorical data? Give an example. [U][4]
9. Name any two sources of data in data science. [K][4]
10. What is meant by structured data? [U][4]

Part B

Short Essay Questions

Answer any four questions. Each question carries 6 marks.

11. A factory was producing electric bulbs of average life length 2000 hours with S.D 300. A new process was introduced with the hope that the life length of bulbs would increase. A sample of 50 bulbs produced by the new process was found to have an average length of life 2200 hours.

Examine whether it is reasonable to think that the length of life of bulbs has increased Assuming that the standard deviation has not changed. [A][1]

12. The following data was obtained in an investigation about the effect of vaccination for small pox:

	Vaccinated	Not vaccinated
Attacked by smallpox	3	12
Not Attacked	8	5

Examine whether the vaccination is effective in preventing smallpox. [A][2]

13. Differentiate between supervised and unsupervised learning with examples. [U][3]
14. Explain logistic regression with a real-world application. [U][3]
15. Discuss the types and classifications of digital data used in data science. [U][4]
16. Explain different types of data. [U][4]

Part C

Essay questions

Answer any one question. Each question carries 12 marks.

17. Explain the procedure of one-way ANOVA [U][2]
18. (i). What is multiple linear regression?

(ii). Derive the normal equations used to estimate the regression coefficients and illustrate its practical use with an example. [A][3]